Chulin ZHAO

Phone: +86 15940441329 Email: 2617944@dundee.ac.uk

Educational Background

Central South University (Hunan, China) | University of Dundee (Dundee, UK) **Sep. 2023- Jul. 2027(Expected)**

Bachelor of Engineering in Computer Science (English-taught) (Joint Program)

GPA: 89.44/100 (Top 20% cohort)

Major Coursework GPA: 94/100

Core Courses: Mathematical Logic; Multi-paradigm Programming and Data Structures; Algorithms and Artificial Intelligence; Object-Oriented Analysis and Design; Computer Networks and Data Communications; Database Systems; Information Security Technology; Agile Software Engineering; Computational Theory

SKILLS

Programming:

- Python, PyTorch, SQL

Machine Learning & AI:

- Representation Learning, Contrastive Learning, Transfer Learning & Fine-Tuning
- Multi-Label Learning, Information Retrieval, Knowledge-Enhanced Modeling
- LLM-based Agent Systems, Clinical NLP

Tools:

- Hugging Face Transformers

Research Interests

Trustworthy AI in Healthcare, Uncertainty Quantification & Calibration, LLM Agents, Generative AI, Bioinformatics

Publication

- Zhao, C., Xu, Y., & Liu, S. (2026). Calibrated Closed-Form Uncertainty for Radiative Gaussian Splatting in Sparse-View CT. (https://doi.org/10.48550/arXiv.2607.13682)
- When Composition Doesn't Add Up: Humans Identifying Defects in AI-Generated Images (2026 IEEE International Workshop on Multimedia Signal Processing, Accepted)

Research Experiences

Closed-Form Uncertainty for Radiative Gaussian Splatting in Sparse-View CT
(Ongoing, https://arxiv.org/abs/2607.13682) **Jun. 2026-Sep. 2026(Expected)**

- Built a variational density posterior on R^2-Gaussian. Exploiting strict linearity in X-ray rendering, closed-form predictive variance is computed exactly in a single forward pass, at substantially lower cost than sampling-based estimators.Derived exact closed-form predictive variance in voxel and projection space, evaluated in a single forward pass without Monte Carlo (enabled by linear X-ray rendering).
- Conducted the first systematic calibration study for Gaussian-splatting CT (Spearman / AUSE / ECE with temperature scaling) on the official 15-scene benchmark across three view budgets: full-volume error ranking passes on 14/15 scenes.
- Proposed CGCA (coverage-gated + closed-form scoring) for active view acquisition: in anisotropic scenes, achieves view selection comparable to a 10-replica ensemble at roughly 1/10 the scoring cost.
- Core diagnosis: posterior variance is a constraint map, not an error map—high-σ regions reliably flag unconstrained tissue (trustworthy alarms), while low-σ regions do not certify accurate reconstruction (all-clears are not trustworthy).

Knowledge-Enhanced Agentic EHR-to-ICD-9 Multi-Label Prediction (Ongoing)
(Independent Research Project) **April. 2026-Present**

- Assigns ICD-9 codes from unstructured EHR notes by aligning structured evidence with knowledge-expanded code text.

- An agent extracts typed clinical evidence (diagnoses, history, procedures, discharge diagnoses, hospital course).
- Each ICD code is expanded along four axes—definition, synonyms, EHR phrasing, and coding boundaries—and refined by an Evaluate–Improve loop: Evaluate diagnoses matching failures from held-out metrics and writes a failure briefing; Improve rereads the briefing and rewrites the four-axis text.
- The resulting representations are used to train a matching model between notes and codes.

Ride the Waves: Exploring Sustainable Tourism Development Options for Juneau
(2025 MCM/ICM Mathematical Contest in Modeling, Problem B Program) **Feb. 2025**

- Led a 3-person team as project lead; decomposed the problem into sub-modules covering data collection, multi-objective modeling, cross-region transfer, sensitivity analysis, and policy recommendation.
- Collected and preprocessed data from EPA, BLS, Alaska Conservation, and Juneau tourism economic reports; standardized variables across different scales for unified modeling.
- Constructed five objective functions around total visitor volume, covering tourism revenue, carbon footprint, infrastructure load, resident satisfaction, and glacier loss; designed composite objective function and constraints balancing economic, environmental, and social targets.
- Applied Genetic Algorithm, PSO, and NSGA-II to explore optimal visitor capacity and resource allocation strategies under varying constraints.
- Extended the model to Hawaii by re-parameterizing for O'ahu and Moloka'i, using PSO for visitor distribution optimization to compare policy priorities across over-tourism and low-traffic regions.
- Conducted sensitivity analysis to identify key drivers including per-capita spending, total visitor volume, and glacier sensitivity coefficient; translated findings into actionable policy recommendations.
- Coordinated team task allocation, modeling decisions, and English paper writing under competition time constraints, ultimately winning the S Award.

When Composition Doesn’t Add Up: Humans Identifying Defects in AI-Generated Images **Jan. 2025-Nov. 2025**

- Co-designed the CO-AID dataset targeting systematic defects in text-to-image models (Midjourney, Imagen, FLUX) under complex compositional prompts involving multiple entities, attributes, spatial relations, and interactions.
- Curated 651 reference images across four categories (people, hand, object, scene); refined compositional prompts through manual revision of ChatGPT-generated descriptions to emphasize multi-entity and relational complexity.
- Designed a two-tier defect annotation taxonomy (global-level and local-level); organized subjective experiment pipeline including training, pilot, and main study phases with 29 participants performing multi-label defect annotation
- Performed data cleaning and statistical analysis on human annotation results; converted participant click locations into defect maps for local defect localization.
- Fine-tuned TranSalNet on defect maps for local defect prediction; integrated predicted maps as spatial guidance into GPT-Image-1 for defect-guided image repair, achieving improved localization alignment with human annotations compared to direct GPT-Image-1 detection.
- Participated in proof-of-concept experiments validating the dataset’s utility for both defect prediction and image repair tasks.

Honors & Awards

- S Award of the 2025 MCM/ICM Mathematical Contest in Modeling 2025
- Third-Place Academic Scholarship, Central South University 2025